Queue order

A queue follows First In, First Out (FIFO). Enqueue inserts at the rear and dequeue removes from the front. Items with equal priority leave in their arrival order in this ordinary queue.

Linear array limitation

In this simple linear array queue, rear only increases and removed cells are not reused until the queue is reset. If rear reaches the final array index, another enqueue fails even if earlier positions became free after dequeues. A circular representation reuses those positions.

Circular array conditions

For this circular queue of capacity 5, front and rear start at -1. Empty means front is -1. Full means (rear + 1) % 5 equals front. On the first enqueue set front and rear to 0. Removing the last remaining item resets both indices to -1.

Deque operations

A deque supports insertion and removal at both its front and rear. This differs from the one-way insertion and removal convention of an ordinary queue. A doubly linked representation with head and tail pointers can perform these four end operations in constant time.

Priority queue policy

In this priority queue, larger numeric priorities are more urgent. Removal chooses the greatest priority rather than the earliest arrival. With priorities 2, 9, and 4, the item of priority 9 is removed first.